More recently, drones have been tested and deployed for organ transportation, where the viability of organs such as kidneys, hearts, and livers depends on strict time and temperature conditions. UAV-based organ transport bypasses road traffic constraints and reduces the risk of delays, increasing the likelihood of successful transplant outcomes. Specialized medical drones equipped with secure, insulated storage and real-time telemetry ensure environmental control, location tracking, and delivery verification.

By combining speed, reliability, and accessibility, drones support life-saving medical logistics, improving emergency responsiveness and enhancing healthcare equity in underserved regions.

Table 8.7 summarizes the diverse roles of UAVs in healthcare logistics, showing how drones serve as rapid, reliable connectors within medical supply networks. By overcoming distance, terrain, and transport delays, UAVs enhance the speed, accessibility, and responsiveness of essential healthcare services, particularly in remote or emergency-affected regions.

Table 8.7 Functional roles of drones in healthcare delivery.

Application area	Primary function	Key benefits	Examples of use
Medical supply transport	Delivery of blood, vaccines, medicines, and diagnostic samples	Reduces delivery time; bypasses traffic and terrain; supports emergency care	Rwanda and Ghana national drone delivery programs (Zipline)
Organ transport	Rapid transport of transplant organs under controlled conditions	Maintains organ viability; increases transplant success rate	Hospital-to-hospital organ relay flights in the United States and India
Mobile laboratory linkage	Transport of lab specimens between remote	Faster diagnosis; improved disease surveillance	Drone-supported tuberculosis and